

AMENDMENT NO. 1 DECEMBER 2023
TO
IS 16090 : 2013 GEO-SYNTHETICS — GEO-TEXTILES USED AS PROTECTION (OR CUSHIONING) MATERIALS — SPECIFICATION

(Page 1, clause 4.3, fourth sentence) — Substitute the following for the existing sentence:

'Polyolefin material shall be UV stabilized by adding suitable UV stabilizer and/or carbon black. Polyolefin material, if manufactured by using carbon black shall contain 2 percent to 3 percent of carbon black by mass with satisfactory dispersion.'

(Page 2, Table 1) — Substitute the following for existing table:

Table 1 Requirements for Geo-textiles Used as Geo-membrane Protection (or Cushioning) Materials

(Clauses 4.4, 4.5, 4.6 and 6.4)

Sl No.	Property	Requirements						Method of Test, Ref to
		Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Mass per unit area, g/m ²	300	400	600	800	1 100	1 200	IS 14716
ii)	Tensile strength, kN/m	16	21	27	32	36	45	IS 16635
iii)	Trapezoidal tear strength, kN	0.42	0.51	0.66	0.89	0.96	1.32	IS 14293
iv)	CBR puncture resistance, kN	3.0	3.5	4.0	5.0	7.5	11.0	IS 16078
v)	Maximum UV resistance (after 500 h exposure), percent	70	70	70	70	70	70	IS 13162 (Part 2)

(Page 4, Annex A) — Substitute the following for the existing entries for IS 13321 (Part 1) : 1992, IS 14716 : 1999

IS No.	Title
IS 13321 (Part 1) : 2022/ ISO 10318-1 : 2015	Geosynthetics: Part 1 Terms and definitions (<i>first revision</i>)
IS 14716 : 2021	Geosynthetics — Test method for the determination of mass per unit area of geotextiles and geotextile-related products (<i>first revision</i>)

(Page 4, Annex A) — Delete the entry for IS 3162 (Part 5) : 1992 and IS/ISO 12236 : 2006.

(Page 4, Annex A) — Insert the following new entries at the end:

IS No.	Title
IS 16078 : 2013/ ISO 12236 : 2006	Geosynthetics — Static puncture test (CBR test)
IS 16635 : 2017/ ISO 10319 : 2015	Geosynthetics — Wide-width tensile test